

A Categorical Naturality Framing of Derivative Asset Replication and Pricing

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The Black Scholes Partial Differential Equation for pricing derivative assets can be recovered and motivated from a naturality condition expressed in category theory. In category theory, Natural Transformations set constraints on two Functors (constrained maps across categories). Since functors map objects to objects and morphisms to morphisms, natural transformations essentially set these mappings in accord with each other, ensuring that there are isomorphic routes from one mapped object to another mapped object. At a high level, Black Scholes is all about isomorphic paths in market space, isomorphic changes in value corresponding to changes in the market. I should get the exact same payoff (and have the same exposure) at every step through time if I own the option, or I own the hedging portfolio. If the paths are not isomorphic, there is opportunity for arbitrage.

In this view, arbitrage traders are rewarded for fixing and maintaining the natural transformation. Though they may not see it this way, market makers are constantly annealing naturality diagrams so they commute.

Here, I want to highlight the structure of a risk-free, no arbitrage replication portfolio—the heart of Black Scholes—as a natural transformation. I am intentionally staying high level, borrowing from category theory its structure, its intuitions, and not necessarily building out a comprehensive and fully sound Categorical Black Scholes Model. The characteristic amenability of category theory to visual diagrams, abstracted objects and arrows, and simple rules like composition and identity make it the perfect tool for highlighting mathematical *structure*.

Further, I will use a one-object monoidal category representing increments on real numbers, $\Delta_{\mathbb{R}}$. Why use a one-object category if we are specifically talking about mapping *specific values* to other *specific values*? The reason follows from the original definition of derivative prices in the Black Scholes PDE. The price of a derivative asset is defined by equal changes in its value with changes in time, underlying asset price, risk free rate, and any other change in the market. How an option and a replication portfolio *change* in value, how they are exposed to multidimensional uncertainty, is in fact what defines the price. When you create derivatives on specific assets—like stocks—then of course absolute values become vitally important and intrinsic to derivative pricing. Prices, in the market, are not truly all the same object! However here we are sketching the structure of *generalized* replications of derivative assets. For this reason, a one-object category trivially ignores *what the object (value, price) is* and instead focuses on aligning *the morphisms on that object*.

The root constraint in Black Scholes is that the change in the derivative's price should always be equal to the change in the replication portfolio's value: $\delta V = \delta \Pi$. We could also say that going long the derivative and short the portfolio (or vice versa) should yield the risk free rate (no nontrivial profit). This defines a portfolio's exposure: $\delta V - \delta \Pi = \delta \Pi - \delta V = 0\%$ (assuming a risk free rate of 0%).

Let's dive in.

We have the category **State** containing pairs of time and asset price, (t, P) , and morphisms, Δ , between them (representing infinitesimal forward steps in time, incurring changes in stock price). The morphism Δ is defined:

$$\Delta(t, P) = (t + \delta t, P + \delta P)$$

δP is more involved than a trivial infinitesimal step, but here we keep it abstract. When we actually solve the PDE for assets defined by Geometric Brownian Motion, we will expand this.

We have functors $\mathbf{F}_{\text{Option}}$ and $\mathbf{F}_{\text{Hedge}}$ that map **State** objects to the category of Real Increments, $\Delta_{\mathbb{R}}$. It contains one object, $\star$, and morphisms of the form $\star \rightarrow \star$. These morphisms represent translations $x \rightarrow x + \delta$. Composition corresponds to addition:

$$(x \rightarrow x + \delta_1) \circ (x \rightarrow x + \delta_2) = x \rightarrow x + (\delta_1 + \delta_2)$$

Again, incredibly abstract, but it is a category of value changes on real numbers. Specifically the functors map all states to the sole object, $\star$,

$$\mathbf{F}_{\text{Option}}(t, P) = \mathbf{F}_{\text{Hedge}}(t, P) = \star$$

And they map morphisms, Δ , in **State** to translation morphisms:

$$\mathbf{F}_{\text{Option}}(\Delta)(x) = x + \delta V$$

And

$$\mathbf{F}_{\text{Hedge}}(\Delta)(x) = x + \delta \Pi$$

Thus these functors represent and preserve how values representing options and hedging portfolios change as we move through state space.

Now we have a natural transformation, $\Theta :: \mathbf{F}_{\text{Option}} \Rightarrow \mathbf{F}_{\text{Hedge}}$, which gives us our constraint. It identifies the value of the option and the value of the hedging portfolio at all times; it is the categorical expression of replication. Its components are defined as follows:

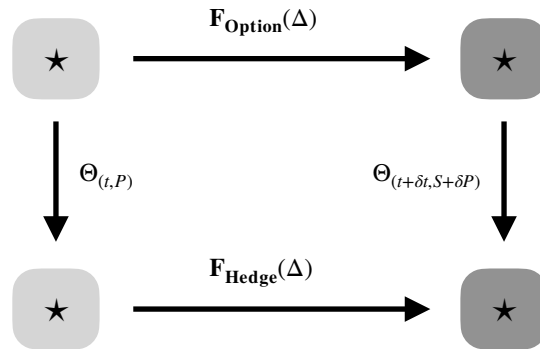
$$\Theta_{(t, P)} = \text{id}_{\star}$$

Practically, it is an identity map, but it is vital in the following commutation diagram that must hold:

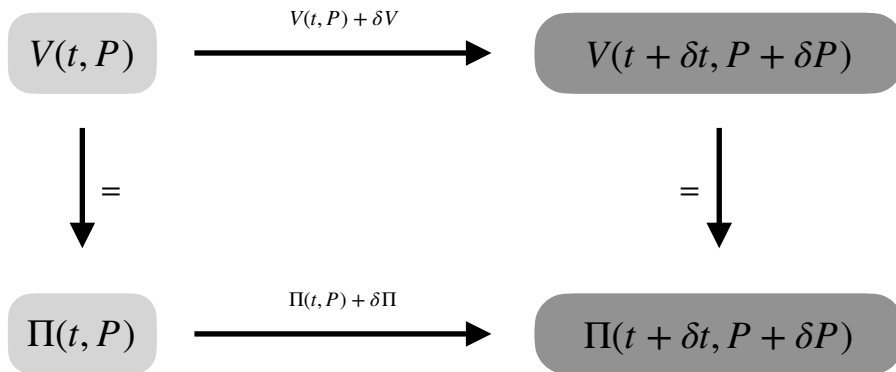
$$\Theta_{(t+\delta t, P+\delta P)} \circ \mathbf{F}_{\text{Option}}(\Delta) = \mathbf{F}_{\text{Hedge}}(\Delta) \circ \Theta_{(t, P)}$$

Essentially, this enforces that both sides (which represent, in our minds, $V(t, P) \rightarrow \Pi(t + \delta t, P + \delta P)$) map reals isomorphically. These functors must map the morphism, Δ , in **State** to the same morphism in $\Delta_{\mathbb{R}}$. This is our no arbitrage constraint: going forward in state space as an option (via $\mathbf{F}_{\text{Option}}(\Delta)$) and then mapping to the hedge value (via $\Theta_{(t+\delta t, P+\delta P)}$) should be the exact same thing as first mapping to the hedge value (via $\Theta_{(t, P)}$) and then going forward in state space as a hedge (via $\mathbf{F}_{\text{Hedge}}(\pi)$).

The fully faithful naturality diagram looks like:



Though, intuitively, we might imagine something (no longer in $\Delta_{\mathbb{R}}$!) more like this:



Because Θ is just an identity map, what this really enforces is:

$$\mathbf{F}_{\text{Option}}(\Delta)(x) = \mathbf{F}_{\text{Hedge}}(\Delta)(x)$$

$$x + \delta V = x + \delta \Pi$$

$$\delta V = \delta \Pi, \quad \forall x$$

Because morphisms in $\Delta_{\mathbb{R}}$ are dependent only on their infinitesimal increments, δ , and not on the inputs, x , the only way two morphisms can be equal is if their increments are the same. Now, we have an equivalence of morphisms in $\Delta_{\mathbb{R}}$. Making another casual jump, we can just talk about real numbers and we can set up the Black Scholes PDE.

Again, I use category theory and natural transformations to demonstrate the structure required to be maintained in this equivalence, $\delta V = \delta \Pi$. If we were being thorough, we'd be in a continuous category that could handle the stochastic calculus we need, we'd be coming not from **State** but instead a category which likewise could have stochastic elements in its morphisms, and so on. That is a worthy pursuit, but it would mask what I am trying to show. Category theory gives us an alternative and beautiful view of the *structure* of the constraint that Black Scholes solves.

Further, I have come to like this monoidal, hands-off approach to model the necessary constraint in a replication portfolio of a derivative asset because it puts the focus in the right place: *change*. The pricing of derivatives (and indeed *the naming*) is entirely defined by exposure to change; monoidal categories are perfect denizens to represent this exposure to change, because what exclusively gives the category structure is the mechanism of change. Any comparison of objects is trivial because there is one; the comparisons and relationships of *morphisms* is what matters. This is the perfect tee-up to a differential equation.

Now that we've seen the structure, we can crunch numbers...